

Multiple scattering in the He-3 capsule wall sets the MX17 opening-angle resolution

A survey of every reconstruction option for the Micromegas-only high-energy measurement,

what 10^7 fully simulated pairs say about the hard limit, and what can still be done

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Internal note — informal. Drafted by Claude (Anthropic AI assistant) under D. Neff’s direction; simulation, numbers and text reviewed by D. Neff.

Part I — Summary

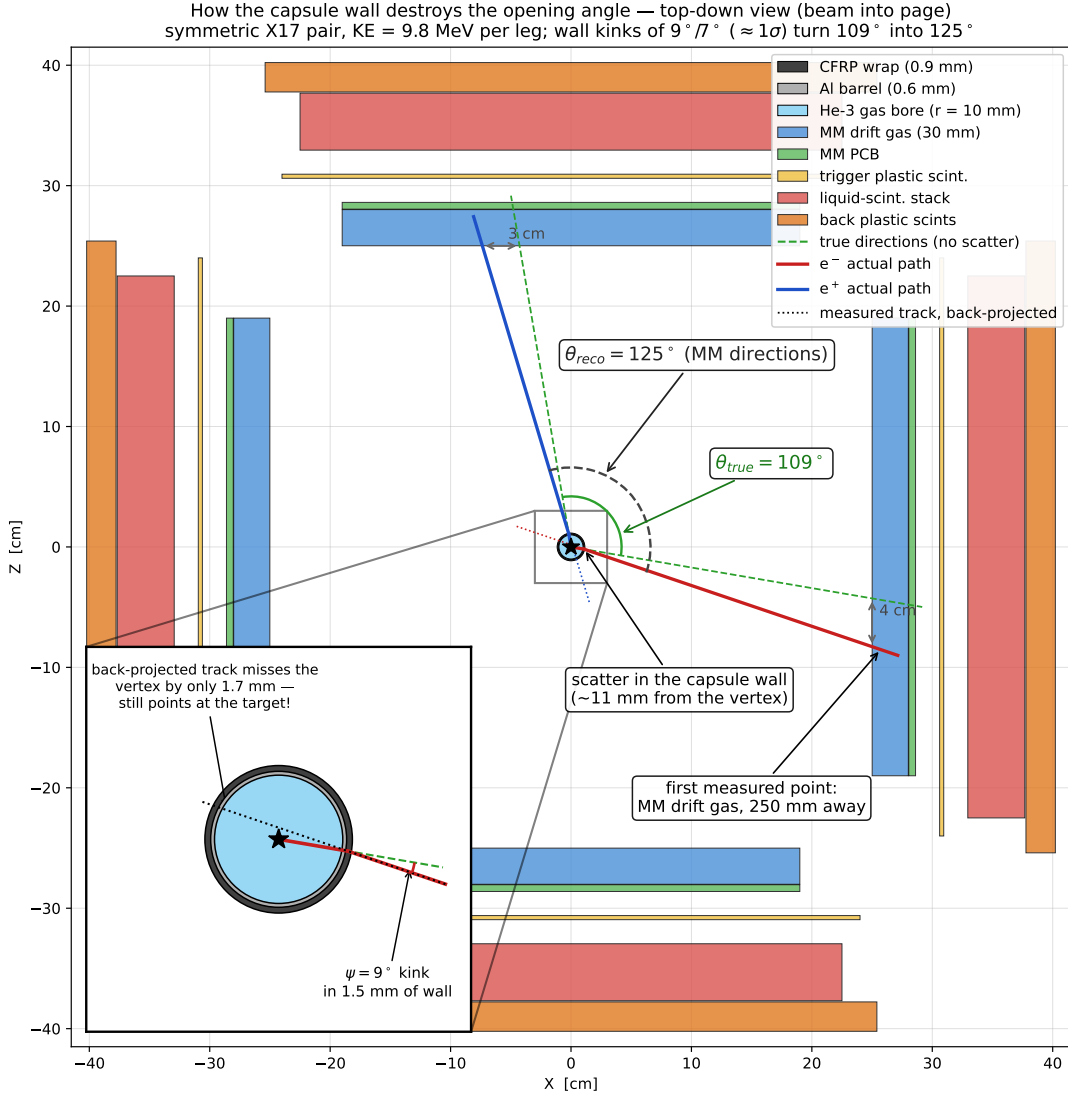
The question

The MeV note (12 June) settled production: ~ 32 X17 produced per day over the full MeV region. With the thermal measurement eliminated, the scintillator triggers are dropped: the DAQ records a fixed $\sim 10\ \mu\text{s}$ readout per pulse, from the γ flash down to $E_n \approx 20\ \text{keV}$ — which covers essentially all of the production — and **every pair in MM geometric acceptance is recorded** (“MM double”, both legs reaching a drift volume, same-arm pairs included: 19.6% for X17, 23.6% for IPC). That gives ~ 6.4 recorded X17/day, ~ 190 per 30-day run, on a recorded signal-to-IPC ratio of ≈ 0.021 . The remaining question is whether those events form a *visible peak*. For this high-energy measurement the calorimeters are taken as saturated by the γ -flash environment and unusable, so the only per-event observable is the e^+e^- **opening angle measured by the Micromegas (MM) trackers**. The at-rest X17 signal is a sharp kinematic shoulder at $\theta_{\min} = 2 \arcsin(m_{X17}/E^*) \approx 109^\circ$; IPC is a smooth continuum. **This note asks: what is the best opening-angle resolution any reconstruction can achieve, is the target-vertex constraint being exploited, and is the resulting resolution good enough?**

What we did

We analysed the 10^7 -event at-rest pair sample (`pairs_v2_step-target`: 5.0×10^6 X17 + 5.0×10^6 IPC, full STEP small-capsule geometry, MM front faces at 250 mm) with *nine* per-particle direction estimators and *seven* pair-level opening-angle methods — including oracle methods that use the true vertex, which bound what any real algorithm could do. We cross-checked the simulation against an analytic Highland multiple-scattering (MS) budget of the real material stack, localized where the scattering happens, and propagated the measured resolution onto the signal shape.

The answer

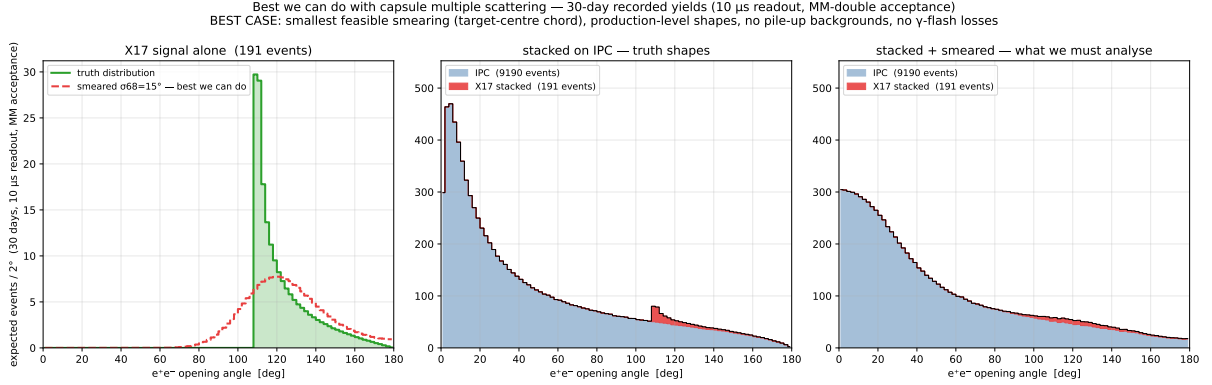


The picture above is the whole result. Both legs of the pair scatter in the capsule wall — 0.6 mm of Al plus 0.9 mm of CFRP, crossed ~ 11 mm from the vertex — by a typical (Highland) $\theta_0 \approx 7^\circ$ per leg at the energies of a symmetric X17 pair. Everything downstream of that wall measures the *kinked* directions perfectly well, and reconstructs the wrong opening angle. The zoom shows why no constraint can repair it: the kink happens so close to the origin that the back-projected track still passes within ~ 2 mm of the vertex — **the scattered event remains fully consistent with “a pair from the target”**. Quantitatively, for the full accepted spectrum ($\sigma_{68} =$ half the 16–84% span of $\Delta\theta = \theta_{reco} - \theta_{truth}$):

pair-level method	σ_{68} X17	σ_{68} IPC
direction at first MM hit (perfect tracker)	15.5°	15.0°
PCA fit of the drift track (+0.5 mm hit smear)	16.0° (16.5°)	15.5° (15.5°)
chord: true vertex $\rightarrow$ first MM hit (<i>oracle</i>)	14.5°	13.0°
chord: target centre $\rightarrow$ first MM hit (<i>realistic</i>)	15.0°	13.5°

All seven methods (the remaining three sit between these) land within 2° of each other, and the *realistic* vertex-constrained chord is within 0.5° of the truth-vertex oracle: with the small capsule, the “pairs come from the target” trick is already implemented and already saturated.

Selecting the most symmetric pairs (min-leg KE > 8 MeV) reaches the plateau $\sigma_{68} \approx 12.5^\circ$ (11.5° oracle) — the floor. The consequence for the measurement:



This is the bottom line of the whole study, in expected 30-day recorded events. **Left:** the truth X17 distribution — the sharp 109° kinematic shoulder — against the *best we can do* given the capsule multiple scattering, smeared by the smallest feasible resolution (the target-centre chord, $\sigma_{68} = 15^\circ$): a $\sim 40^\circ$ -FWHM bump at a quarter of the height. **Middle:** that truth signal stacked on the recorded IPC continuum — even unsmeared it is only a $\sim 60\%$ local excess. **Right:** both species smeared — *this is the spectrum the analysis actually has to extract the X17 signal from*: a $\lesssim 17\%$ broad enhancement of a featureless falling continuum. And this is a *best-case* picture: smallest feasible smearing, no MM acceptance shaping, no pile-up backgrounds, no γ -flash losses. **A cut-and-count peak search on the opening angle is dead, with or without reconstruction improvements.** The resolution is frozen by the wall: it contributes 79% of the $x/X_0 = 1.26\%$ upstream budget, the simulation reproduces the analytic Highland prediction, and the scattering-localization study places the damage at the wall radius. The MM spatial resolution is irrelevant (a 0.5 mm hit smear costs 0.5°).

What can still be done

In order of leverage (Part II §7–9):

1. **Fit the spectrum, don't cut it.** The response matrix $P(\theta_{\text{reco}} | \theta_{\text{truth}})$ is measured with 10^7 events; a template likelihood fit (smeared signal + IPC continuum + pile-up backgrounds from the Stage-4 sampler) converts resolution into a statistics penalty instead of a veto. The go/no-go number becomes the expected significance of that fit with ~ 190 signal events per 30 days — the natural next study, and the machinery exists.
2. **MM-internal quality cuts as an energy proxy.** The $30\text{--}40^\circ$ tails come from legs below ~ 3 MeV. Without calorimetry they can still be tagged inside the MM (track straightness, drift-gas dE/dx , range), buying the $15.5 \rightarrow 12.5^\circ$ plateau.
3. **Bias correction** ($+3.5^\circ$ at high KE, larger for soft pairs) — free, automatic in a response-matrix fit.
4. **Hardware** is the only way below 12° : removing the Al barrel (53% of the budget) scales $\sigma \propto \sqrt{x/X_0}$ to $\approx 0.7\times$, i.e. $\sigma_{68} \approx 10^\circ$. Fig. 6 (§8) shows this does *not* qualitatively change the picture.

The bottom line: the opening-angle resolution is hardware-limited at $\sigma_{68} \approx 12\text{--}15^\circ$ by ~ 1.5 mm of capsule wall sitting 11 mm from the vertex, every reconstruction trick including the target-vertex constraint is already within $\sim 1^\circ$ of its theoretical bound, and the X17 angular peak does not survive as a visible bump. If the experiment is recoverable, it is recoverable at the spectrum-fit level, not the event level — quantifying that fit is the follow-on study this note motivates.

Part II — The full story

1 The measurement this note is about

The DAQ records a fixed $\sim 10\mu\text{s}$ readout per pulse starting at the EAR2 γ flash (no scintillator trigger; the 0.2–2 MeV core of the production arrives 1.0–3.2 μs in, and the readout reaches $E_n \approx 20\text{ keV}$). The working assumption for this note — pending the flash-recovery measurement — is that the liquid-scintillator calorimetry is saturated and contributes nothing event-by-event. What remains per pair is geometry: two charged tracks in the MM arms. From two directions one can form exactly one discriminating quantity, the opening angle θ . For a decay $X17 \rightarrow e^+e^-$ the invariant mass obeys

$$m^2 = 2m_e^2 + 2(E_+E_- - p_+p_- \cos \theta), \quad (1)$$

so with energies unmeasured, θ carries all the mass information we will ever get. For an at-rest capture ($E^* = 20.58\text{ MeV}$) the X17 signal populates $\theta \geq \theta_{\min} = 2\arcsin(16.8/20.58) = 109.4^\circ$, with a sharp shoulder at the threshold (symmetric pairs) and a tail to 180° (asymmetric pairs). IPC fills a smooth continuum. The experiment’s discriminating power is therefore set by one number: how well θ is reconstructed. For scale: near the shoulder, $dm/d\theta \approx (E^*/2)\cos(\theta/2) \approx 0.10\text{ MeV/deg}$, so a 15° angular error is the mass-equivalent of $\pm 1.6\text{ MeV}$ — but only for symmetric pairs; for the rest, the unknown energy sharing already smears the $\theta \leftrightarrow m$ map before the detector does anything.

2 Anatomy of a measured pair

The summary diagram (Part I) shows one worked event in the true top-down geometry: a symmetric X17 pair (9.8 MeV kinetic energy per leg), vertex at the capsule centre, one leg toward the $+X$ arm and one toward the $+Z$ arm. Each leg crosses $\sim 10\text{ mm}$ of 500-bar ^3He , then 0.6 mm of aluminium and 0.9 mm of CFRP, then 24 cm of air, and is first *measured* in the MM drift gas at 250 mm. The wall crossing deflects each leg by a random angle of Highland width $\theta_0 \approx 7^\circ$ (at this energy; much worse for softer legs). In the diagram the kinks are drawn at their actual $\approx 1\sigma$ values, 9° and 7° : the measured opening angle becomes 125° instead of 109° , and the hits arrive 3–4 cm away from where the unscattered tracks would have landed.

Figure 1 repeats the identical true pair under three throws of the scattering dice. The same 109° event is reconstructed at 97° , 121° , or 134° — a spread comparable to the full width of the physical signal region. Figure 2 shows the same fact as an ensemble: the 68%/95% cones each track can occupy after the wall, and the distribution that one single true angle becomes. Already this idealized toy — two legs, Gaussian $\theta_0 = 7.1^\circ$ each, nothing else — gives $\sigma_{68} = 9.8^\circ$. The full Geant4 answer for such symmetric pairs is $11.5\text{--}12.5^\circ$ (non-Gaussian MS tails, oblique wall crossings, drift-gas scattering), growing to $14.5\text{--}15.5^\circ$ over the whole accepted spectrum because one leg of an X17 pair is always below 9.8 MeV and $\theta_0 \propto 1/p$.

3 The estimator catalogue

Nine per-particle direction estimators and seven pair-level methods were implemented in `scripts/analyze_pairs.py` and evaluated on the 10^7 -event sample. Per particle, ψ is the space angle between the estimated direction and the truth direction at the vertex:

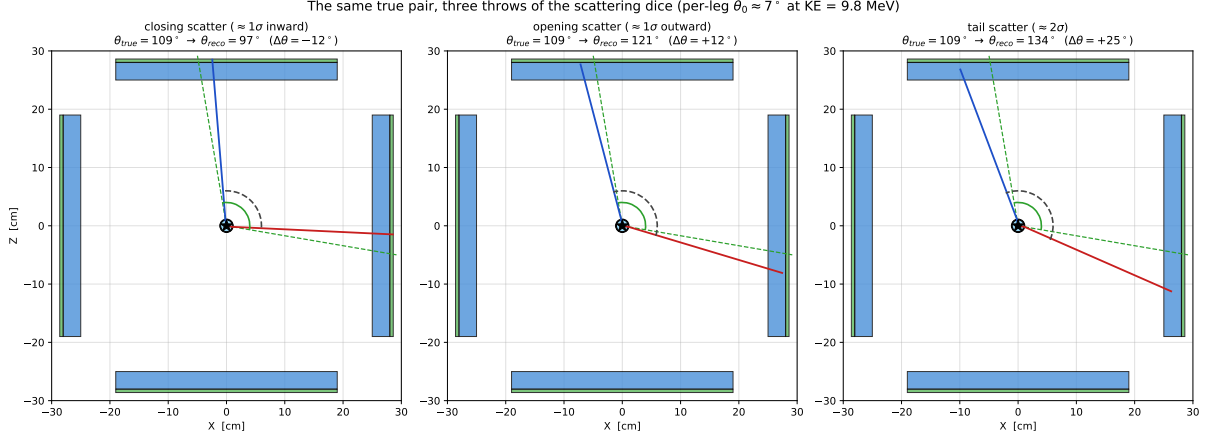


Figure 1: One true pair ($\theta_{\text{true}} = 109^\circ$, symmetric), three scattering outcomes drawn in the real geometry (MM drift gas in blue; downstream stack omitted for clarity). Green dashed: unscattered paths. The wall kink is invisible at the capsule scale — see the Part-I zoom — but dominates the reconstructed angle 250 mm later.

estimator	what it tests	med. ψ (all KE)	KE > 8 MeV
dir. at 1st MM hit	upstream MS only — perfect tracker	10.4°	7.9°
dir. at last MM hit / 1st-vs-last	adds / isolates drift-gas MS	(small; PDF §13)	
PCA fit of drift hits	realistic track fit	10.4°	7.9°
PCA fit vs 1st-hit dir.	fit-method error alone	($\ll$ MS; PDF §13)	
chord: true vtx $\rightarrow$ 1st hit	oracle vertex constraint	9.4°	7.1°
chord: true vtx $\rightarrow$ edep centroid	centroid vs first-hit choice	9.4°	7.1°
chord: target centre $\rightarrow$ 1st hit	realistic vertex constraint	9.9°	7.9°
chord: vtx $\rightarrow$ trig-scint hit ($\sigma=3\text{cm}$)	scintillator fallback	11.4°	9.4°

and at pair level (σ_{68} of $\Delta\theta$, full accepted sample; overall RMS for the first method is $19.1^\circ/20.6^\circ$ X17/IPC):

pair method	X17	IPC
dir. at 1st MM hit (current default)	15.5°	15.0°
PCA drift-track fit	16.0°	15.5°
PCA fit + 0.5 mm hit smear	16.5°	15.5°
chord: true vertex $\rightarrow$ 1st hit (oracle)	14.5°	13.0°
chord: true vertex $\rightarrow$ drift centroid	14.5°	13.5°
chord: target centre $\rightarrow$ 1st hit	15.0°	13.5°
chord: vertex $\rightarrow$ trigger-scint hit	15.5°	15.0°

Three structural facts stand out. (i) The PCA fit gains nothing over the single first-hit direction: averaging the drift hits removes fit noise but integrates drift-gas scattering, and the two cancel. (ii) A 0.5 mm hit smear — a pessimistic MM cluster resolution — costs only 0.5° : **detector granularity is irrelevant**; even mm-scale resolution would not move the result. (iii) Every estimator converges to the same upstream-MS floor (Fig. 3); the floor itself tracks the analytic Highland curve of the material budget.

4 Where the angle is lost: the material budget

The plane-projected Highland width for a singly charged particle crossing x/X_0 radiation lengths is

$$\theta_0 = \frac{13.6 \text{ MeV}}{\beta_{cp}} \sqrt{x/X_0} [1 + 0.038 \ln(x/X_0)], \quad (2)$$

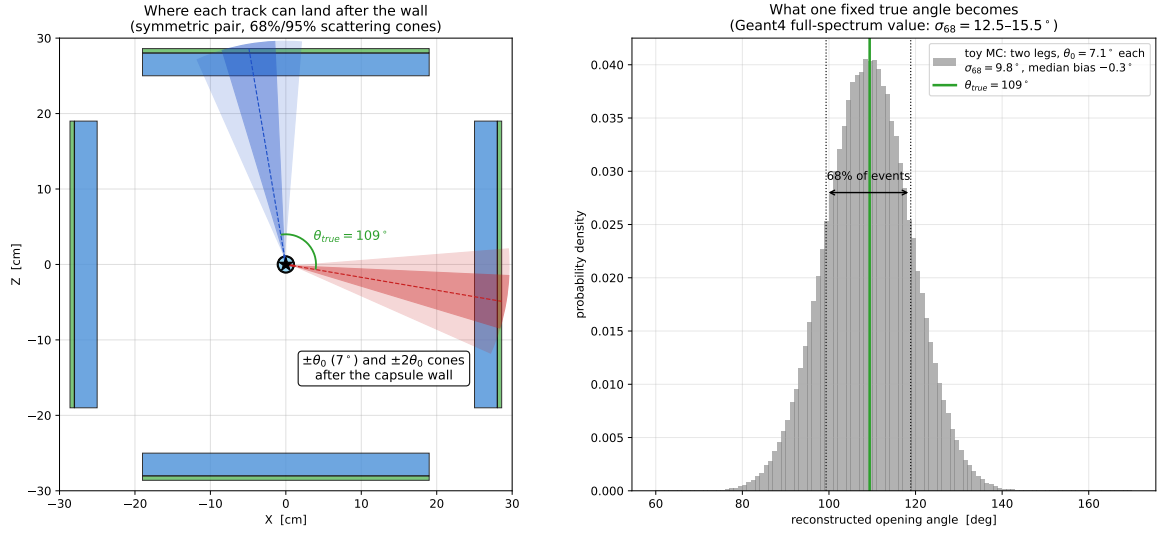


Figure 2: Left: the 68%/95% cones available to each leg of a symmetric pair after the capsule wall. Right: the opening-angle distribution that a single fixed 109.4° pair becomes in a toy MC with only the wall scattering ($\theta_0 = 7.1^\circ$ per leg, Gaussian) — $\sigma_{68} = 9.8^\circ$ before any detector effect. The full simulation gives $12.5\text{--}15.5^\circ$.

and the median *space* angle for Gaussian MS is $\sqrt{2\ln 2}\theta_0 \approx 1.18\theta_0$. Evaluating Eq. 2 layer by layer along the radial exit path (Fig. 4):

layer	x/X_0	share	θ_0 at $p = 10\text{ MeV}/c$
^3He gas, 500 bar, $\sim 10\text{ mm}$ path	0.089%	7%	1.7°
Al barrel, 0.6 mm	0.674%	53%	5.2°
CFRP wrap, 0.9 mm	0.327%	26%	3.5°
air gap, $\sim 23.9\text{ cm}$	0.078%	6%	1.6°
MM window + cathode (Mylar/Kapton/Cu)	0.094%	7%	$\leq 1.4^\circ$
total upstream	1.26%		7.3°

The measured first-hit direction error at $\text{KE} \approx 10\text{ MeV}$ (7.9° , Fig. 3) sits on this prediction, and the scattering-localization study in the analysis PDF (effective single-scatter distance $s_{\text{eff}} = t - \delta / \tan \psi$) independently places the median scatter at the wall radius. So the chain of evidence closes: the resolution is the wall, the wall is $\sim 80\%$ of the budget, and simulation and analytic model agree. Note what is *not* on the list: the 500-bar gas itself (7%), the air (6%), and every MM material ($\leq 7\%$ combined). Pressure is innocent; the *pressure vessel* is guilty.

5 Why the vertex constraint is already exhausted

The target-as-vertex idea is geometrically sound and is implemented (the “target centre $\rightarrow$ first hit” chord). The reason it buys so little is worth stating precisely, because it also rules out every variant of the idea.

(a) **The chord inherits the scatter.** A kink ψ at distance s from the vertex displaces the track at the measurement plane L by $(L - s) \tan \psi$; the chord from the vertex to that point therefore makes an angle $\psi(L - s)/L$ with the true initial direction. With $s \approx 11\text{ mm}$ and $L = 250\text{ mm}$, the chord retains 0.956ψ . That 4% rebate is the entire $\sim 1^\circ$ oracle improvement seen in the tables.

(b) **Moving the tracker closer makes it worse.** The chord error $\rightarrow 0$ as $L \rightarrow s$, so a measurement plane just outside the wall plus a vertex would recover the true direction — but

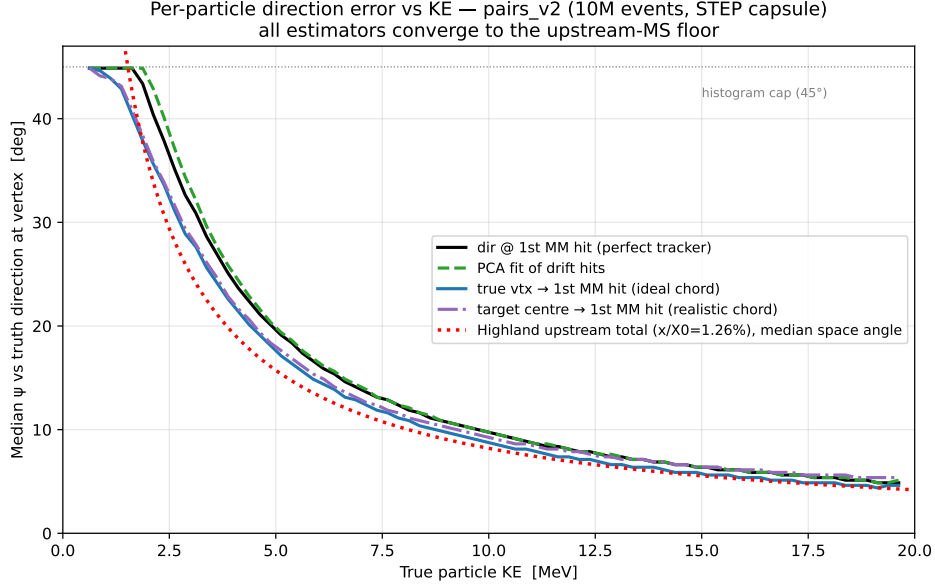


Figure 3: Median per-particle direction error vs kinetic energy for the four exported estimators, against the Highland prediction for the full upstream budget. Below ~ 2 MeV the histogram saturates at its 45° cap — such legs are effectively directionless.

the vertex is unknown within the capsule (RMS ≈ 16 mm along the beam, ≈ 5 mm transverse), adding σ_v/L to the chord. At $L = 250$ mm that costs $\sim 3^\circ$ (measured: the realistic chord is 3.1° in quadrature above the oracle); at $L = 50$ mm it would cost $> 10^\circ$. Short lever arms lose more to vertex ignorance than they gain on the scatter; the current 250 mm is already near the optimum.

(c) The event cannot tell you its own vertex. Back-projected tracks miss the true vertex by a median of 1.0 cm (KE 10–15 MeV) to 3.7 cm (3–4 MeV) — at or beyond the 10 mm bore radius. A two-track vertex fit is therefore *weaker* than the prior “somewhere in the capsule”; equivalently, the Part-I zoom shows the scattered track still pointing at the target. Pointing cuts keep their value for background rejection, but carry no information to correct the angle with.

Together (a)–(c) bound every vertex-flavoured scheme: chords, constrained fits, vertex refits, extra tracking layers. With the small capsule the realistic chord sits 0.5° from the truth-vertex oracle, and the oracle sits 1° from the perfect tracker. **Total theoretical headroom from smarter geometry-based reconstruction:** $\lesssim 1.5^\circ$.

6 What cannot work at all without calorimetry

A kinematic fit is the standard rescue when a constraint is unused, so we checked the counting explicitly. Unknowns per event: the two momentum magnitudes (directions are measured). Available at-rest-capture constraints: energy conservation ($E_+ + E_- = 20.58$ MeV) and the fixed parent momentum ($|\vec{p}_+ + \vec{p}_-| = 11.9$ MeV/c). Imposing *both* solves for the two magnitudes — but then $m^2 = (E_+ + E_-)^2 - |\vec{p}_+ + \vec{p}_-|^2$ is fixed to 16.8 MeV identically, independent of the data: the fit returns its own hypothesis. Imposing only one leaves one equation for two unknowns. There is no middle ground: **without an energy measurement there is no per-event mass and no kinematic sharpening of θ .** The opening angle, as smeared by the wall, is the final observable.

7 What still can be done, with expected gains

1. **Template fit of the θ spectrum** (the main path). The measured response $P(\theta_{\text{reco}}|\theta_{\text{truth}})$, including its bias and non-Gaussian tails, is in `analysis/pairs_v2/geant4_response.json`

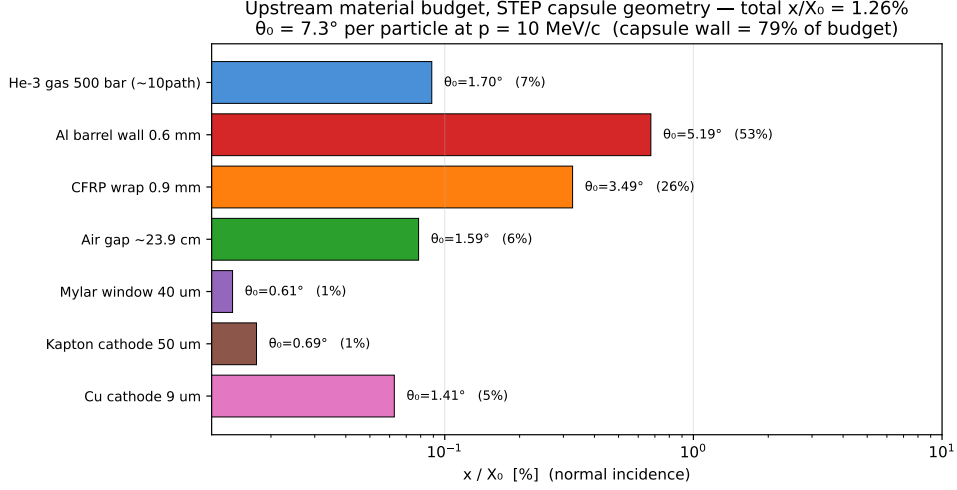


Figure 4: Upstream material budget for the STEP capsule geometry.

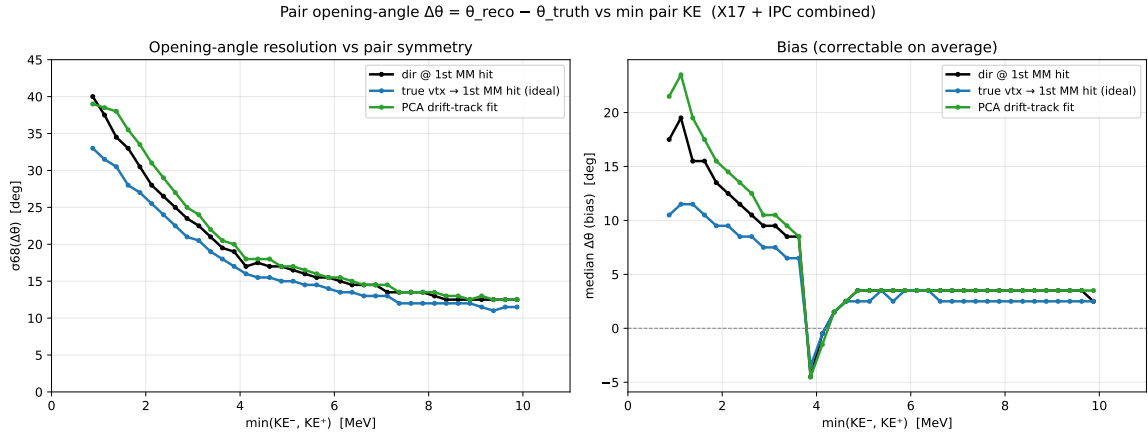


Figure 5: Resolution and bias vs the softer leg’s kinetic energy. The plateau above ~ 8 MeV is the symmetric-pair floor; the soft-leg events supply the long tails. (The bias spike at ~ 4 MeV is a quantile/binning artifact of a bimodal residual, not physics.)

and already loads into the `MX17_Simulation` fast-MC. A binned likelihood fit of signal+background templates uses the full shape information; resolution degrades the *statistical power* smoothly instead of erasing the peak. The required follow-on study: expected significance vs run time, with the IPC continuum and the Stage-4 pile-up backgrounds under the realistic response. Until that number exists, the experiment is neither dead nor alive.

2. **MM-internal soft-leg rejection.** Fig. 5: cutting the soft legs moves $15.5^\circ \rightarrow 12.5^\circ$ and removes the worst tails (and the worst bias). The study above used *truth* KE; the open task is to build the proxy from MM-only quantities (drift dE/dx , track straightness, range/penetration) and measure its efficiency–purity. Soft legs are also the most-scattered legs, so the proxy tags exactly what we want to remove.
3. **Bias correction:** $+2.5\text{--}3.5^\circ$ on the plateau, tens of degrees for soft pairs — absorbed automatically by item 1.
4. **Statistical micro-gains:** per-event vertex- y estimate from the two hit positions, optimal chord+direction combination — $\lesssim 0.5^\circ$ combined, bounded by §5.

8 The hardware ceiling

If a future capsule iteration is on the table, $\sigma \propto \sqrt{x/X_0}$ sets the exchange rate:

change	upstream x/X_0	σ scale	full-spectrum σ_{68}
current (Al 0.6 mm + CFRP 0.9 mm)	1.26%	1.00	15.5°
halve both walls	0.76%	0.78	12°
remove Al barrel (all-composite vessel)	0.59%	0.68	10.5°
remove the wall entirely (unphysical)	0.26%	0.45	7°

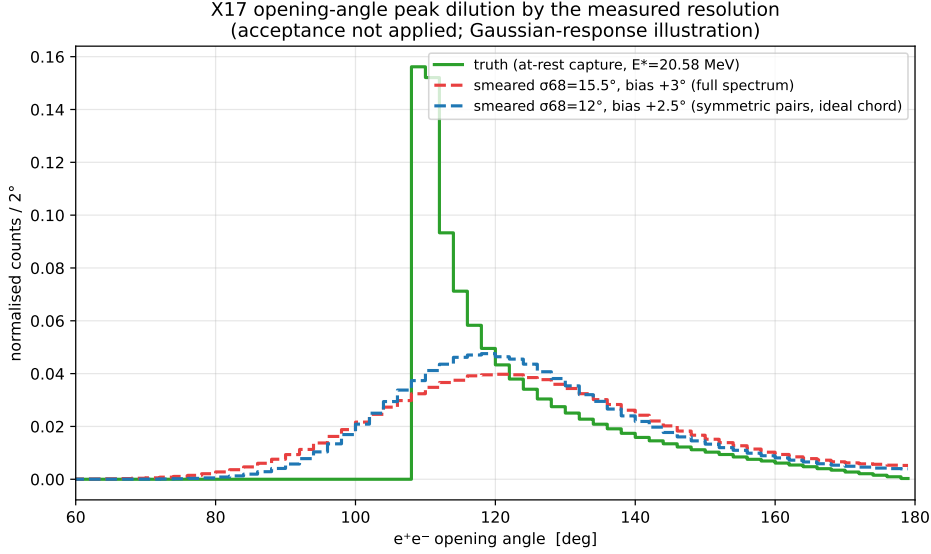


Figure 6: The X17 truth spectrum against two smearing scenarios: the current full-spectrum resolution ($\sigma_{68} = 15.5^\circ$, red) and the symmetric-pair/ideal-chord floor ($\sigma_{68} = 12^\circ$, blue) — which is also where the realistic hardware options of the table land. The two smeared curves are nearly indistinguishable: improving the resolution by the maximum plausible amount barely changes what is observable.

Even the unphysical limit leaves a 7° smearing of a shoulder whose physical width is comparable — and the realistic options land at $10\text{--}12^\circ$, which Fig. 6 shows is visually indistinguishable from the present situation. Hardware helps the template fit; it does not resurrect the visible peak.

9 Consequences for the X17 search

Putting the pieces together for a 30-day run under the $10\ \mu\text{s}$ readout: ~ 190 recorded X17 pairs whose 109° shoulder is smeared into a $\sim 40^\circ$ -FWHM enhancement, on a recorded IPC continuum ~ 48 times larger before angle cuts ($S/B \approx 0.021$ — the trigger-free readout records the small-angle IPC bulk that the old double-arm trigger silently removed, so the background grows $\times 9$ while the signal grows $\times 2.3$), plus pile-up backgrounds to be quantified by the Stage-4 sampler. The discriminating shape survives only statistically.

Figure 7 and the money plot in Part I put the two species on one axis. Three observations. (i) The IPC continuum is nearly invariant under the resolution; the entire information loss is the signal shoulder. (ii) At truth level the stacked signal would stand $\sim 60\%$ above the continuum at the shoulder (money plot, middle panel); after smearing the maximum local excess is $\sim 17\%$, spread over $\sim 40^\circ$ (right panel — the spectrum the analysis must work with). (iii) Naive counting on these best-case shapes in a $105\text{--}150^\circ$ window gives $S \approx 135$ on $B \approx 950$ ($S/\sqrt{B} \approx 4.4$ per 30

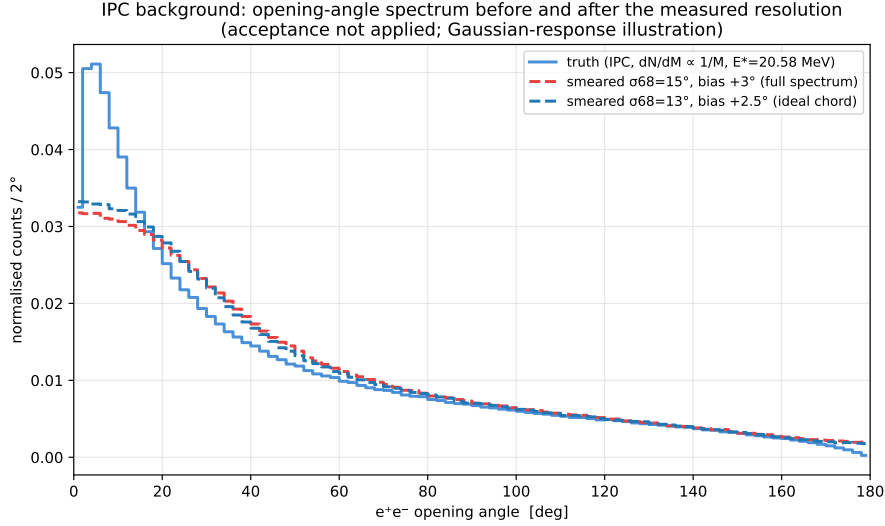


Figure 7: The IPC background spectrum before and after the measured resolution (same construction as the X17 dilution plot in Part I; the virtual-state mass is sampled from the generator’s $dN/dM \propto 1/M$). The one structure IPC has — the small-angle pile-up from low-mass pairs — is flattened, and *everything above $\sim 60^\circ$ is essentially invariant*: a smooth continuum smeared by 15° is the same smooth continuum. The resolution damages only the signal.

days, down from 6.2 unsmeared) — superficially healthy, but three things eat into it: an excess of $S/B \approx 0.14$ is only readable if the IPC normalisation (α_{IPC} , currently an *assumed* 2.1×10^{-3}) is fit simultaneously from the off-window shape, so the result rides on how well the large IPC sample constrains its own shape; the MM acceptance vs θ is not applied (small-angle same-arm pairs must also be *resolved* as two tracks in one chamber — pairs that merge into one track drop out of the background, pairs that survive sit at small θ , away from the signal); and the non-IPC pile-up backgrounds are absent here. The right framework is therefore the simultaneous shape fit of §7 item 1, and the decision-grade question this note hands to the next study is concrete: *given the measured response matrix, the at-rest (later boosted) signal template, and the simulated background spectrum, what signal significance does a binned likelihood fit deliver per 30 days?* If that number is $\gtrsim 3\sigma$, the measurement survives as a statistical shape analysis; if it is $\ll 1\sigma$, no reconstruction or plausible vessel redesign will change the verdict — the conclusion would be that an MM-only opening-angle measurement of X17 at this geometry is not viable, and attention should turn to recovering some calorimetric information (e.g. flash-recovery measurements, or triggering schemes that protect part of the LS dynamic range).

10 Caveats

- **At-rest kinematics.** MeV-window captures form $^4\text{He}^*$ at $E^* = 20.58 + \frac{3}{4}E_n$ MeV with a CM boost: the shoulder shifts and softens by a few degrees and the energy sharing changes. The resolution numbers are material-driven and robust; the *templates* need the E_n -dependent generator (open item) before a final fit.
- **Gaussian illustration.** The money/dilution plots smear with a Gaussian of the measured $\sigma_{68} + \text{bias}$; the real response has heavier tails ($\text{RMS}/\sigma_{68} \approx 1.25$), which slightly worsens the realistic picture. The template-fit study must use the full response matrix, not the Gaussian.
- **Soft-leg medians** below ~ 2 MeV saturate the 45° histogram cap; quoted values there are lower bounds (the conclusion is unaffected — those legs are directionless either way).

- **Yield inputs.** Production 32.4 X17/day ($E_n \gtrsim 20$ keV, MeV-note decades) and MM-double acceptances 19.6%/23.6% (X17/IPC, pairs-v2), with the inherited assumptions $\alpha_{\text{IPC}} = 2.1 \times 10^{-3}$ and X17/IPC = 2.5%. The 10 μ s-readout scheme also assumes the MMs themselves recover from the γ flash early enough — the flash-recovery measurement applies to the trackers, not only the calorimeters.

Appendix: reproduction

- Underlying analysis (lxplus): `scripts/analyze_pairs.py` over the 100 `pairs_v2` STEP-target files on EOS, producing — in `analysis/pairs_v2/` — the 17-section `pair_analysis_v2.pdf` (estimator definitions §12–16) and `geant4_response.json` (ψ tables, σ_{68} /bias validation, energy response).
- Figures in this note: `python scripts/make_angular_report_figures.py` (local venv); shares code with `scripts/make_angular_resolution_figs.py`, which builds the PNG versions for `docs/angular_resolution/angular_resolution_note.md` (the working mark-down version of this note).
- Numbers: per-particle medians and pair σ_{68} from the analysis PDF legends and the response JSON; Highland budget computed in the figure script from `DetectorConstruction.cc` dimensions (bore $r = 10$ mm, Al 0.6 mm, CFRP 0.9 mm, MM at 250 mm, ^3He at 500 bar, $\rho = 62.7$ mg/cm³).
- Build: `cd docs/report && pdflatex angular_note.tex` (run twice for references).